Exercise Sheet 2

Probability, Data and Estimation

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer



2.A Conditioning and Bayes

Exercise 2.A.1: Consequences of the Axioms

Using only the three Kolmogorov axioms (and set algebra), show:

1. $P(A \setminus B) = P(A) - P(A \cap B)$
2. $P(A \cup B) \leq P(A) + P(B)$
3. If $A \subseteq B$, then $P(A) \leq P(B)$.

Exercise 2.A.1: Consequences of the Axioms — Solution

Exercise 2.A.1: Consequences of the Axioms — Solution

1. $A = (A \setminus B) \cup (A \cap B)$, disjoint, so by Axiom III $P(A) = P(A \setminus B) + P(A \cap B)$.
2. $A \cup B = A \cup (B \setminus A)$, disjoint: $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$, since $P(A \cap B) \geq 0$ (Axiom I).
3. $B = A \cup (B \setminus A)$, disjoint: $P(B) = P(A) + P(B \setminus A) \geq P(A)$ by Axioms III and I.

Exercise 2.A.2: One Flag, Two Flags

Fraud occurs in **0.2%** of transactions. A rule flags **95%** of frauds and **2%** of legitimate transactions.

1. What fraction of flagged transactions is fraud?
2. A second, different rule has the same rates. Assume the two rules err **independently given the class**. A transaction is flagged by both. Now?
3. When would the assumption in 2 be badly wrong — and in which direction would the answer then be off?

Exercise 2.A.2: One Flag, Two Flags — Solution

Exercise 2.A.2: One Flag, Two Flags — Solution

1. $P(\text{flag}) = 0.002 \cdot 0.95 + 0.998 \cdot 0.02 = 0.02186$, so $P(\text{fraud} \mid \text{flag}) = 0.0019 / 0.02186 \approx 8.7\%$.
2. In odds: prior odds $0.002 / 0.998 \approx 0.0020$; each flag multiplies them by the likelihood ratio $0.95 / 0.02 = 47.5$. Two flags: $0.0020 \cdot 47.5^2 \approx 4.52$, i.e. $P \approx 4.52 / 5.52 \approx 82\%$.
3. If both rules look at the same signal (say, both react to foreign IPs), the second flag adds little new evidence, yet the formula counts it twice — the answer is **too confident**. This is exactly the assumption naive Bayes makes.

Exercise 2.A.3: Independent Is Not Disjoint

1. Show that two disjoint events A, B with $P(A), P(B) > 0$ are never independent.
2. Flip two fair coins. Let A = "first is heads", B = "second is heads", C = "the coins agree". Show that A, B, C are pairwise independent but not mutually independent.

Exercise 2.A.3: Independent Is Not Disjoint — Solution

Exercise 2.A.3: Independent Is Not Disjoint — Solution

1. $P(A \cap B) = 0 \neq P(A)\,P(B) > 0$.
2. Each event has probability $\frac{1}{2}$. $A \cap B = \{HH\}$, $A \cap C = \{HH\}$, $B \cap C = \{HH\}$: each has probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ — pairwise independent. But $A \cap B \cap C = \{HH\}$ has probability $\frac{1}{4} \neq \frac{1}{8}$. Knowing A and B determines C.

2.B Random Variables and the Joint

Exercise 2.B.1: App Use and Churn

The joint distribution of X (uses the app: 1/0) and Y (churns: 1/0):

	$Y = 0$	$Y = 1$
$X = 0$	0.30	0.20
$X = 1$	0.40	0.10

1. Compute $P(X = 1)$, $P(Y = 1)$, $P(Y = 1 \mid X = 1)$ and $P(Y = 1 \mid X = 0)$.
2. Are X and Y independent? Compute $\text{cov}(X, Y)$.
3. Does the table show that the app *reduces* churn? What else could produce it?

Exercise 2.B.1: App Use and Churn — Solution

Exercise 2.B.1: App Use and Churn — Solution

1. $P(X{=}1) = 0.5$, $P(Y{=}1) = 0.3$, $P(Y{=}1 \mid X{=}1) = 0.1/0.5 = 0.2$, $P(Y{=}1 \mid X{=}0) = 0.2/0.5 = 0.4$.
2. Not independent ($0.2 \neq 0.3$). $\mathrm{cov}(X, Y) = E[XY] - E[X]E[Y] = 0.10 - 0.5 \cdot 0.3 = -0.05$.
3. No. A joint distribution shows association, not direction: loyal customers may be the ones who install the app (the abduction exercise 1.C.1 of Exercise Sheet 1). Deciding needs an experiment or a model of the causes.

Exercise 2.B.2: Dependent, Yet Uncorrelated

Let X take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $Y = X^2$.

1. Compute $E[X]$, $E[Y]$ and $\mathrm{cov}(X, Y)$.
2. Show that X and Y are not independent.
3. What does this say about using correlation to screen features?

Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

1. $E[X] = 0, E[Y] = \frac{2}{3}, E[XY] = E[X^3] = 0$, so $\operatorname{cov} = 0$.
2. $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = \frac{1}{3}$.
3. A feature can determine the target completely and still have zero correlation with it. Correlation screens only for (monotone) linear association.

2.C Data and Statistics

Exercise 2.C.1: Scale and Model

For each attribute of a customer table, give the scale and a first probabilistic model, and name one statistic that would be **invalid**: payment method (card / invoice / PayPal) · delivery rating (1–5 stars) · number of orders last year · order value in € · has a newsletter subscription.

Exercise 2.C.1: Scale and Model — Solution

Exercise 2.C.1: Scale and Model — Solution

attribute	scale	first model	invalid
payment method	nominal	categorical (p_1, p_2, p_3)	mean, median
rating	ordinal	categorical (order unused) or ordinal model	mean (gaps unknown) — defensible only with care
orders	ratio (count)	Poisson, or binomial with a maximum	— (all valid)
order value	ratio, positive, skewed	log-normal	none, but the mean is outlier-sensitive
newsletter	nominal, binary	Bernoulli(p)	— (coded 0/1, the mean *is* $\hat{p}$)

Exercise 2.C.2: How Precise Is a Conversion Rate?

A campaign reached **400** users; **60** converted.

1. Estimate the conversion rate and its standard error.
2. Give an approximate 95% confidence interval and state correctly what the 95% refers to.
3. How many users would give a half-width of ± 0.01?

Exercise 2.C.2: How Precise Is a Conversion Rate? — Solution

Exercise 2.C.2: How Precise Is a Conversion Rate? — Solution

1. $\hat{p} = 60/400 = 0.15; \mathrm{SE} = \sqrt{0.15 \cdot 0.85/400} \approx 0.018$.
2. $0.15 \pm 1.96 \cdot 0.018 \approx [0.115, 0.185]$. 95% of intervals built **by this procedure** contain the true rate — not "95% probability that the rate lies in this interval".
3. $1.96\sqrt{0.1275/n} = 0.01 \Rightarrow n \approx 4{,}900$ — twelve times the data for a 3.5 times narrower interval: precision costs $\sqrt{n}$.

Exercise 2.C.3: A Sampling Distribution by Simulation (Python)

```python
import numpy as np
rng = np.random.default_rng(0)
samples = rng.exponential(scale=2.0, size=(1000, 25))   # 1000 samples, n = 25
```

1. Compute the mean of each sample and the standard deviation of those means. Compare with $\sigma/\sqrt{n}$ (for this distribution $\sigma = 2$).
2. The exponential is strongly skewed. Plot a histogram of the means — what shape do you see, and which theorem predicts it?
3. Repeat with $n = 100$. By what factor does the spread shrink?

Exercise 2.C.3: A Sampling Distribution by Simulation — Solution

Exercise 2.C.3: A Sampling Distribution by Simulation — Solution

```
means = samples.mean(axis=1)
print(means.std(ddof=1), 2.0 / np.sqrt(25))   # ≈ 0.40 vs 0.40
```

1. About 0.40 against $\sigma/\sqrt{n} = 2/5 = 0.40$.
2. Close to a bell, slightly right-skewed — the central limit theorem, not yet perfect at $n = 25$ for a skewed population.
3. Four times the data, half the spread: about 0.20.